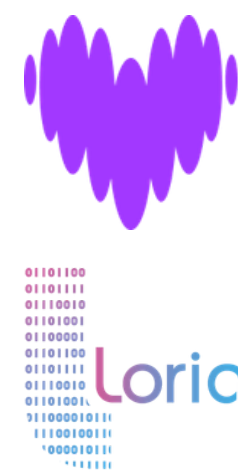


Discovery Quality and the Value of Familiarity



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1 Motivation

Recommendation is typically optimized for a one-time relevance experience, effective in domains with high content volatility (e.g., e-commerce, news).

In **music consumption**, users often **re-listen** to new items they like several times, indicative of satisfaction.

Music familiarity is a key component of discovery and preference formation. Surprisingly, few prior works incorporate it into their modeling.

2 Discovery Quality

We consider:

Weak discovery: song encountered but no subsequent engagement.

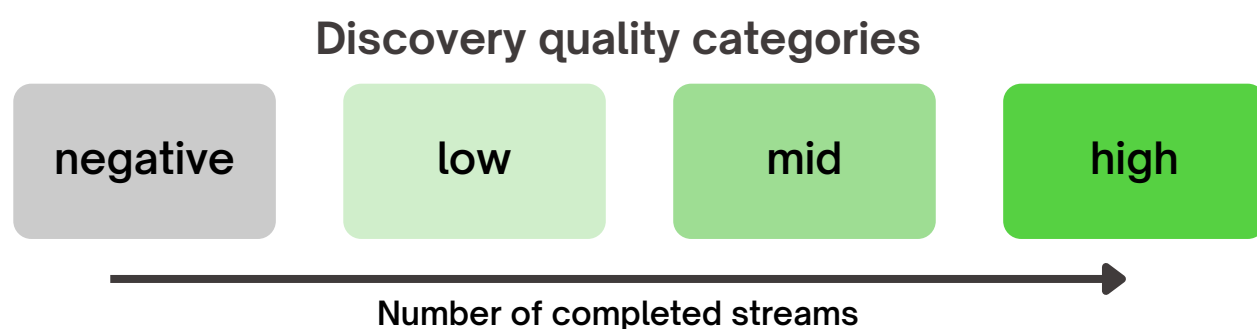
Good discovery: new song encountered → repeated consumption.

Discovery event: sequence of completed streams by a user of an unheard song, starting from its first exposure.

Within a 7-day window:

- Positive: ≥ 2 completed streams;
- Negative: < 2 completed streams.

For positive events, **discovery quality** is defined as the resulting level of user engagement, measured by the number of completed streams.



3 Familiarity measures

We compute an artist level familiarity profile composed of familiarity measures for each user using long-term streaming logs (6 months).

I. Artist Affinity: $\text{Affinity}(u, a) = \frac{\text{LT}_u(a)}{\text{LT}_u^{\text{total}}}$

II. Artist Base-Level Activation: $B(u, a) \approx \ln \left[\sum_{i=1}^k \frac{1}{\sqrt{t_i}} + \frac{2(n-k)}{\sqrt{t_n} + \sqrt{t_k}} \right]$

III. Artist Popularity: $\text{Pop}_{\text{attn}}(a) = |U_a|$

IV. Track Release Date

These familiarity measures values are converted into 5 categories based on their quantiles.

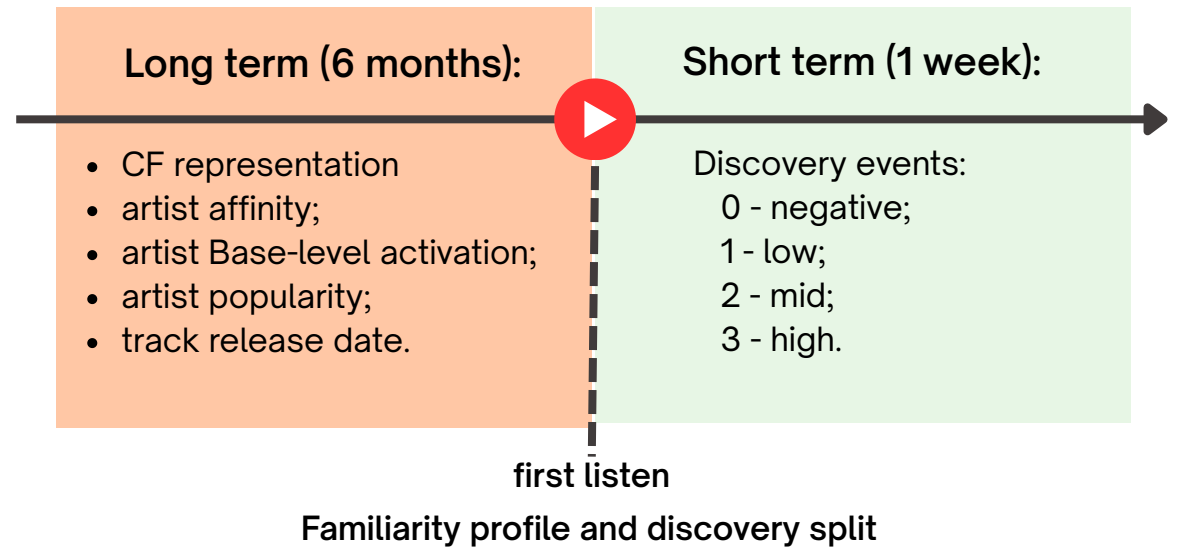
4 Discovery Quality from Familiarity

We investigate whether these familiarity measures are informative for predicting discovery quality through a classification task.

Data: 50K users' streaming data.
Familiarity: 6-month long-term history;
discovery: 1 week. discovery window.



Online repository

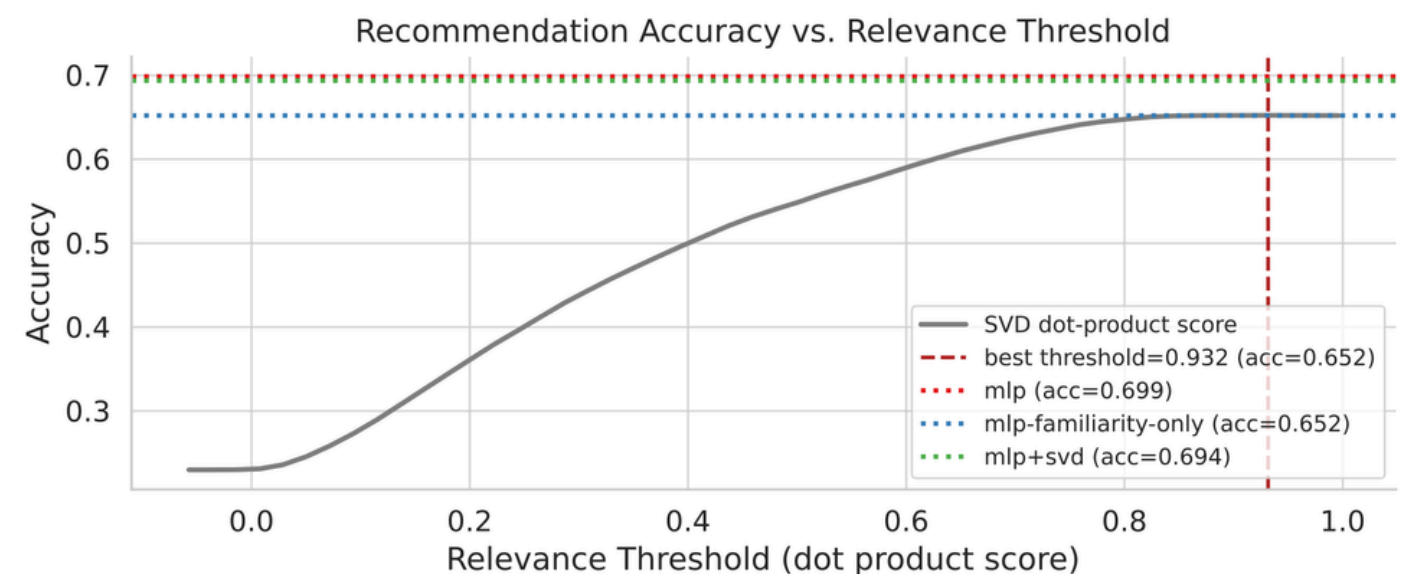


5 Results

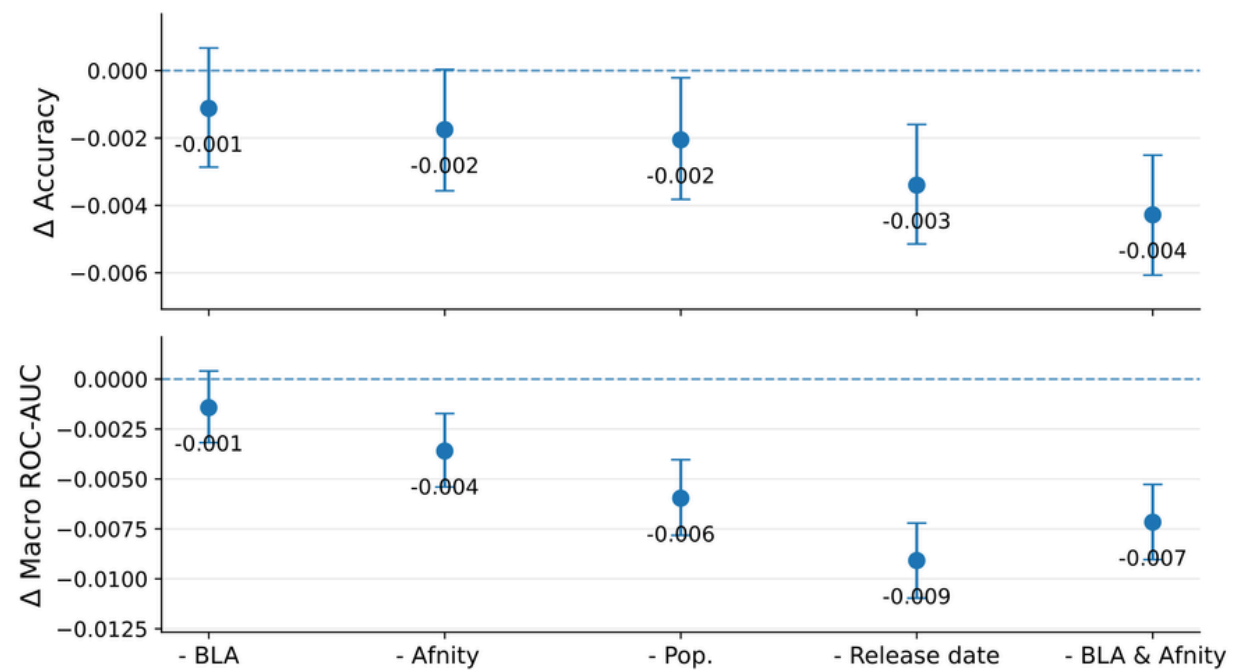
I. Predictability of discovery quality

Model	AUC	F1	Acc
MLP-FAMILIARITY-ONLY	0.5968	0.1974	0.6521
MLP	0.7397	0.3562	0.6986*
MLP+SVD	0.7372	0.3516	0.6937

II. Discovery quality on SVD embeddings



III. Individual contribution of familiarity signals to discovery quality prediction



6 Conclusion

Artist familiarity is a useful signal for predicting discovery quality, complementing user and item representations, with artist popularity and release date contributing most.

Pretrained SVD embeddings can also capture high-quality discoveries, but are less effective than models trained directly for discovery prediction.

Beyond discovery itself, familiarity may help capture whether a new item leads to sustained engagement.